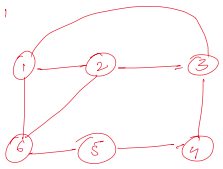
Hamiltonian Cycle

Sunday, 15 February 2026 2:44 PM

(*) If a graph is given then we have to start from some starting vertex and visit all the vertices exactly once & return back to the starting vertex.

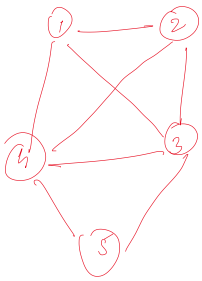
Ex: 1



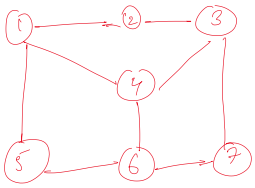
There are some cycle.

1,2,3,4,5,6,1
1,2,6,5,4,3,1
1,6,2,5,4,3,1
2,3,4,5,6,1,2

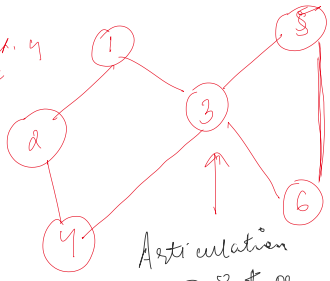
Ex: 2



Ex: 3

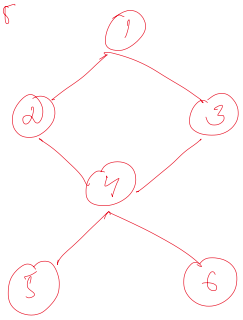


Ex: 4

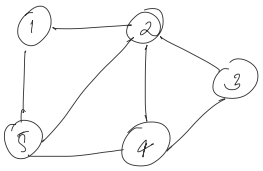


Articulation Point or junction point

Ex: 5



Pendent vertex.



$G_7 =$

	1	2	3	4	5
1	0	1	1	0	1
2	1	0	1	1	1
3	1	1	0	1	0
4	0	1	1	0	1
5	1	1	0	1	0

x

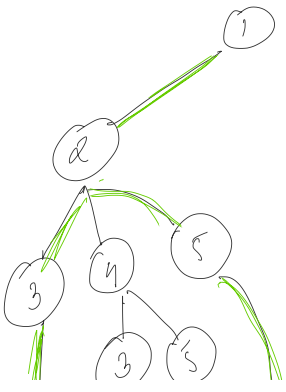
1	2	3	4	5
1	2	3	4	5

(*) Bounding function

(1) no duplicates

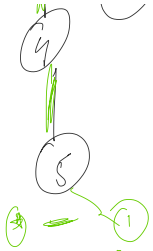
(2) there should be an edge to previous vertex

(3) Last vertex should be edge to first



```
Algorithm Hamiltonian(k)
{
  do
  {
    NextVertex(k);
    if(x[k]==0)
      return;
    if(k==n)
      Print(x[1:n]);
    Hamiltonian(k+1);
  }while(true);
}
```

Algorithm NextVertex(k)



cycle is found

≡ cycle.

```
{
  do
  {
    x[k] = (x[k] + 1) mod (n+1);
```

next value

```
    if (x[k] == 0) return;
    if (G[x[k-1], x[k]] != 0)
```

(A) check if ≡ edge is there to previous

```
    {
      for j=1 to k-1 do if (x[j] == x[k])
                          break;
      if (j == k)
      if ((k < n or (k == n)) && G[x[n], x[1]] != 0
```

(B) check for duplicate

last position

(A) If edge back to vertex 1.

```
        return;
      }
    } while (true);
  }
```